

# **Advanced NLP Tasks**

# Introduciton

# Introduction

**Information extraction (IE)** is the task of **automatically extracting structured information from unstructured** and/or **semi-structured** machine-readable **documents** and other electronically represented sources.

# Introduction

As NLP evolves, so do IE tasks. Traditional tasks evolve, and new ones emerge out of necessity.

What are the most common IE tasks, and what are their related tasks?

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- 1. Named Entity Recognition (NER)**
  - a. Part-of-Speech Tagging (POS)
  - b. Conditional Random Field (CRF)
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- 4. Natural Language Inference (NLI)**
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# Named Entity Recognition (NER)

# Part-of-Speech Tagging (POS)

Named entity recognition (**NER**), aims at **identifying real-world entity mentions from texts**, and **classifying them** into **predefined types**.

Example:

"Suxamethonium infusion rate and observed fasciculations."

"Suxamethonium chloride (Sch) was administered i.v."

# Part-of-Speech Tagging (POS)

We wish to predict an output vector  $\mathbf{y} = (y_1, y_2, \dots, y_L)$ , of random variables, given an observed characteristic vector

$$\mathbf{x} = (x_1, x_2, \dots, x_L)$$

$\mathbf{y}$  takes its value from a list of  $N$  possible values.



# Part-of-Speech Tagging (POS)

POS is the process of mapping words in a text with a label corresponding to their grammatical class.

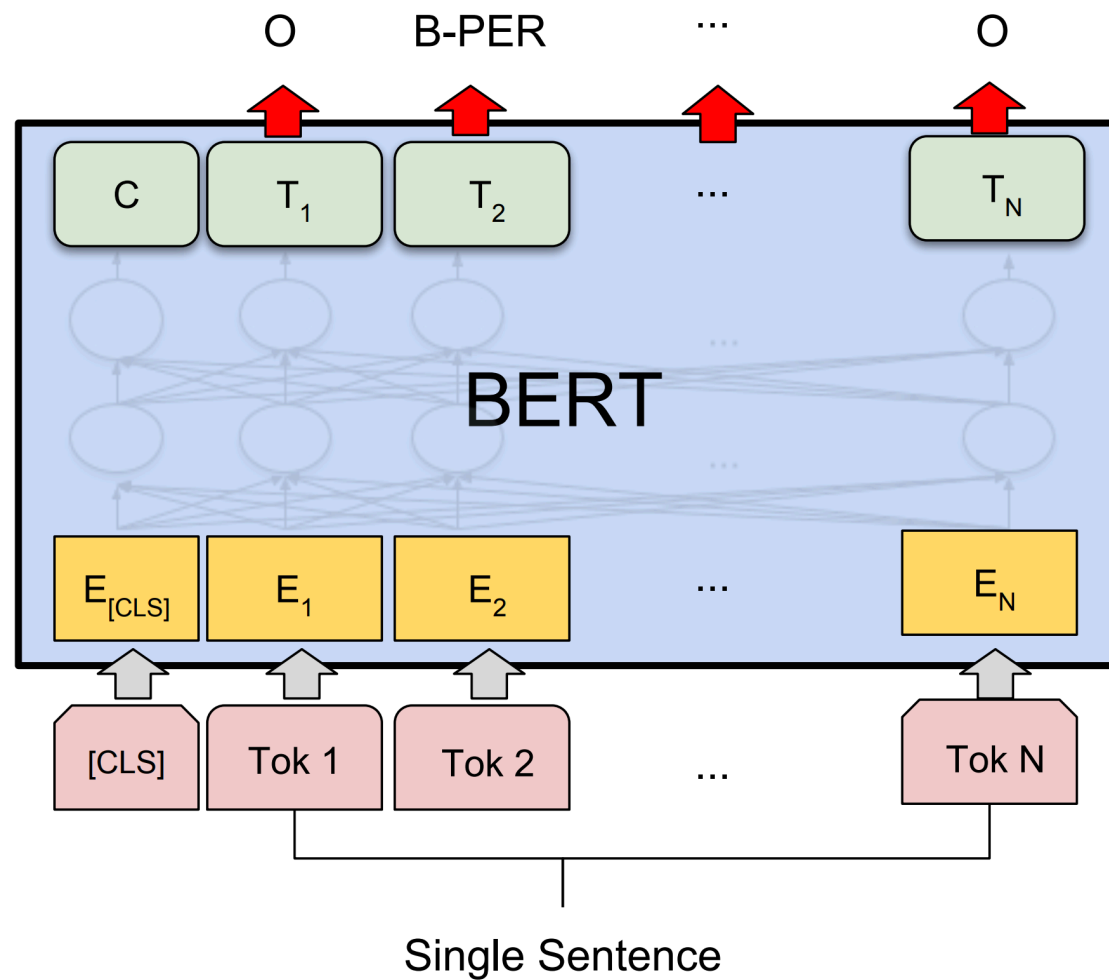
("He", "likes", "to", "drink", "tea"), → ("PERSONAL PRONOUN", "VERB", "TO", "VERB", "NOUN").

# Part-of-Speech Tagging (POS)

There are several levels of granularity.: using [the tag set for english](#)

("He", "likes", "to", "drink", "tea"), → ("PRP", "VBP", "TO", "VB", "NN").

# Conditional Random Field (CRF)



# Conditional Random Field (CRF)

For each token in a sentence at position  $l$  we want to compute a probability  $p$  to belong to a class  $n$ .

$$p : f(\mathbf{x}, \theta)_l \mapsto ?$$

with  $p \in [0, 1]$

# Conditional Random Field (CRF)

Using the softmax function?

$$p : f(\mathbf{x}, \theta)_l \mapsto \frac{e^{f(\mathbf{x}, \theta)_l^{(n)}}}{\sum_{n'=1}^N e^{f(\mathbf{x}, \theta)_l^{(n')}}}$$

The probability given by the softmax function will not encode non-local dependencies!

# Conditional Random Field (CRF)

We need to take sequential decisions: what if we add transition scores into our softmax?

$$p : f(\mathbf{x}, \theta)_l \mapsto \frac{e^{f(\mathbf{x}, \theta)_l^{(n)} + t(y_l^{(n)}, y_{l-1})}}{\sum_{n'=1}^N e^{f(\mathbf{x}, \theta)_l^{(n')} + t(y_l^{(n')}, y_{l-1})}}$$

But this is the probability for one token to belong to a class, we want to compute the probability of a whole sequence of label at once...

# Conditional Random Field (CRF)

$$\begin{aligned} P(\mathbf{y}|\mathbf{x}) &= \prod_{l=2}^L p(\mathbf{y} | f(\mathbf{x}, \theta)_l) \\ &= \prod_{l=2}^L \frac{e^{f(\mathbf{x}, \theta)_l^{(n)} + t(y_l^{(n)}, y_{l-1})}}{\sum_{n'=1}^N e^{f(\mathbf{x}, \theta)_l^{(n')} + t(y_l^{(n')}, y_{l-1})}} \end{aligned}$$

# Conditional Random Field (CRF)

$$\begin{aligned} P(\mathbf{y}|\mathbf{x}) &= \frac{\exp[\sum_{l=2}^L (f(\mathbf{x}, \theta)_l^{(n)} + t(y_l^{(n)}, y_{l-1}))]}{\sum_{n'=1}^N \exp[\sum_{l=2}^L (f(\mathbf{x}, \theta)_l^{(n')} + t(y_l^{(n')}, y_{l-1}))]} \\ &= \frac{\exp[\sum_{l=2}^L (U(\mathbf{x}, y_l^{(n)}) + T(y_l^{(n)}, y_{l-1}))]}{\sum_{n'=1}^N \exp[\sum_{l=2}^L (U(\mathbf{x}, y_l^{(n')}) + T(y_l^{(n')}, y_{l-1}))]} \\ &= \frac{\exp[\sum_{l=2}^L (U(\mathbf{x}, y_l^{(n)}) + T(y_l^{(n)}, y_{l-1}))]}{Z(\mathbf{x})} \end{aligned}$$



# Conditional Random Field (CRF)

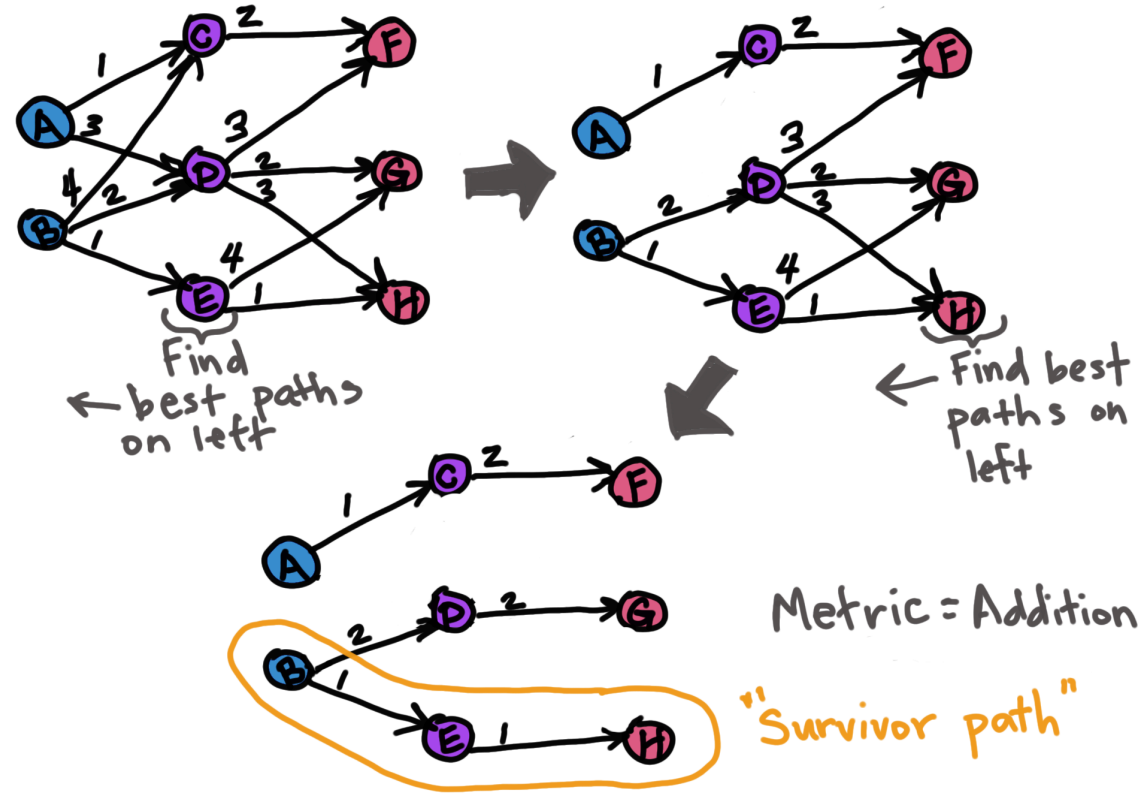
$Z(\mathbf{x})$  is commonly referred as the partition function. However, its not trivial to compute: we'll end up with a complexity of  $\mathcal{O}(N^L)$ .

Where  $N$  is the number of possible labels and  $L$  the sequence length.

How do we proceed?

# Conditional Random Field (CRF)

## Viterbi Algorithm



# Conditional Random Field (CRF)

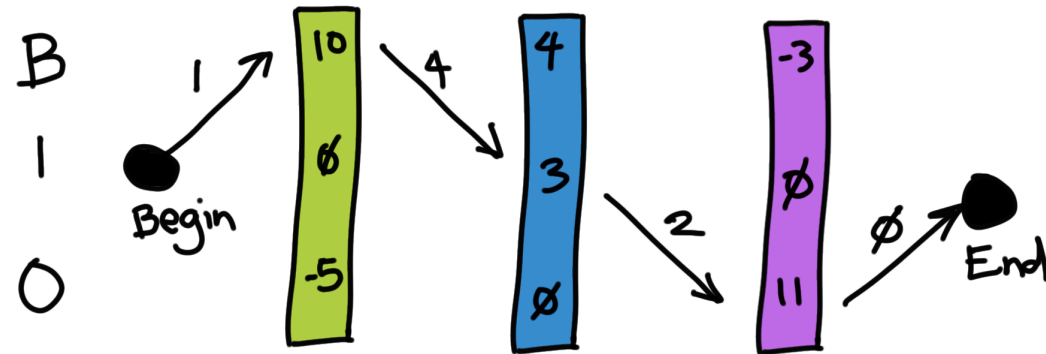
## NER Transition Matrix

|   | B                    | I                    | O                    |
|---|----------------------|----------------------|----------------------|
| B | $C(B \rightarrow B)$ | $C(B \rightarrow I)$ | $C(B \rightarrow O)$ |
| I | $C(I \rightarrow B)$ | $C(I \rightarrow I)$ | $C(I \rightarrow O)$ |
| O | $C(O \rightarrow B)$ | $\infty$             | $C(O \rightarrow O)$ |

$C$  = cost function  
 $\infty$  = wouldn't happen

# Conditional Random Field (CRF)

Linear-Chain CRF Decoded



Python comments help

Best path:  $B \rightarrow I \rightarrow O$

Best score:  $1 + 10 + 4 + 3 + 2 + 11 + 0 = 31$

# Conditional Random Field (CRF)

Negative log-likelihood:

$$\begin{aligned}\mathcal{L} &= -\log(P(\mathbf{y}|\mathbf{x})) \\ &= -\log\left(\frac{\exp[\sum_{l=2}^L (U(\mathbf{x}, y_l^{(n)}) + T(y_l^{(n)}, y_{l-1}))]}{Z(\mathbf{x})}\right) \\ &= -[\log(\exp[\sum_{l=2}^L (U(\mathbf{x}, y_l^{(n)}) + T(y_l^{(n)}, y_{l-1}))]) - \log(Z(\mathbf{x}))] \\ &= \log(Z(\mathbf{x})) - \sum_{l=2}^L (U(\mathbf{x}, y_l^{(n)}) + T(y_l^{(n)}, y_{l-1}))\end{aligned}$$

# Conditional Random Field (CRF)

There is an effective way to compute  $\log(Z(\mathbf{x}))$  with a complexity of  $\mathcal{O}(L)$  using [the Log-Sum-Exp trick](#).

$$\begin{aligned}\log(Z(\mathbf{x})) &= \log\left(\sum_{n'=1}^N \exp\left[\sum_{l=2}^L (U(\mathbf{x}, y_l^{(n')}) + T(y_l^{(n')}, y_{l-1}))\right]\right) \\ &= c + \log\left(\sum_{n'=1}^N \exp\left[\sum_{l=2}^L (U(\mathbf{x}, y_l^{(n')}) + T(y_l^{(n')}, y_{l-1})) - c\right]\right)\end{aligned}$$

# Conditional Random Field (CRF)

If we fix

$c = \max\{U(\mathbf{x}, y_l^{(1)}) + T(y_l^{(1)}, y_{l-1}), \dots, U(\mathbf{x}, y_l^{(N)}) + T(y_l^{(N)}, y_{l-1})\}$   
we ensure that the largest positive exponentiated term is  $\exp(0) = 1$ .

# Sentiment Analysis

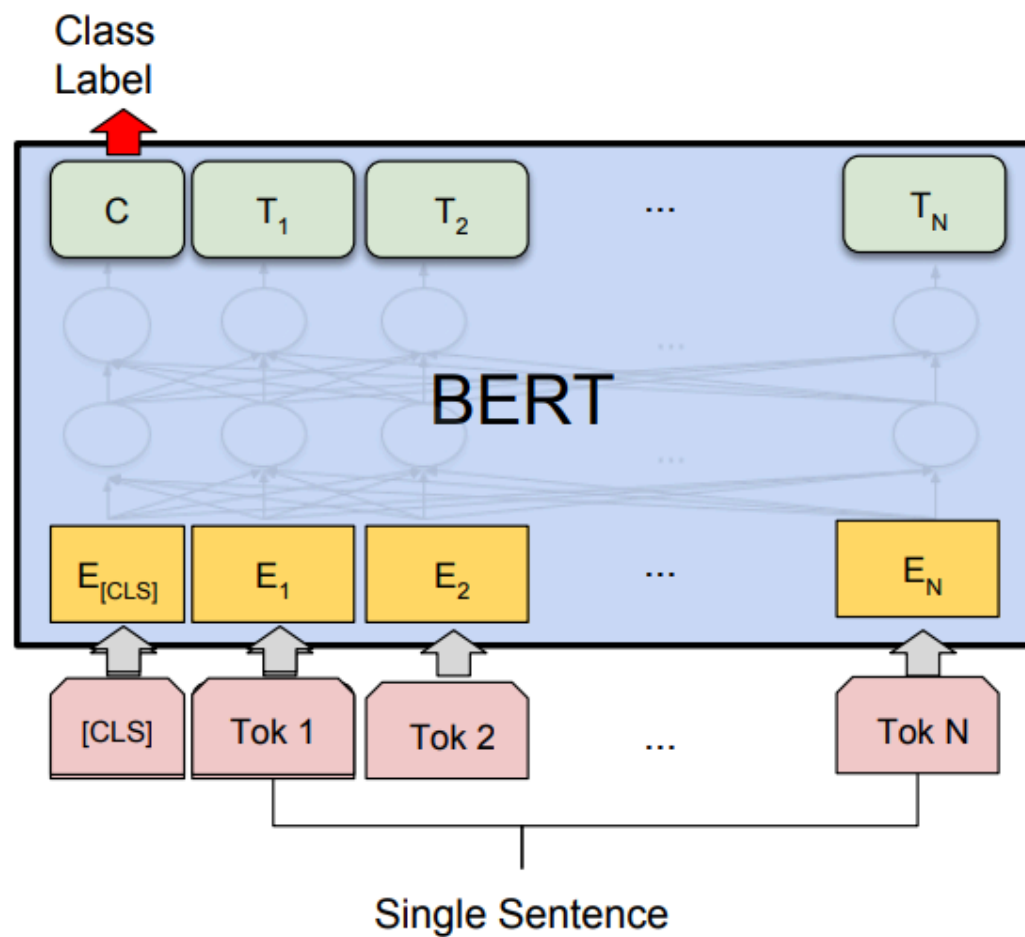


# Sentiment Analysis

**Sentiment analysis** is a sentence classification task aiming at **automatically mapping data to their sentiment**.

It can be **binary** classification (e.g., positive or negative) or **multiclass** (e.g., enthusiasm, anger, etc)

# Sentiment Analysis



# Sentiment Analysis

The loss can be the likes of cross-entropy (CE), binary cross-entropy (BCE) or KL-Divergence (KL).

$$\mathcal{L}_{CE} = -\frac{1}{N} \sum_{n'=1}^N y^{(n)} \cdot \log(f(\mathbf{x}, \theta)^{(n)})$$

$$\mathcal{L}_{BCE} = -y^{(n)} \cdot \log(f(\mathbf{x}, \theta)^{(n)}) + (1 - y^{(n)}) \cdot (1 - f(\mathbf{x}, \theta)^{(n)})$$

$$\mathcal{L}_{KL} = \frac{1}{N} \sum_{n'=1}^N y^{(n)} \cdot \log\left(\frac{y^{(n)}}{f(\mathbf{x}, \theta)^{(n)}}\right)$$

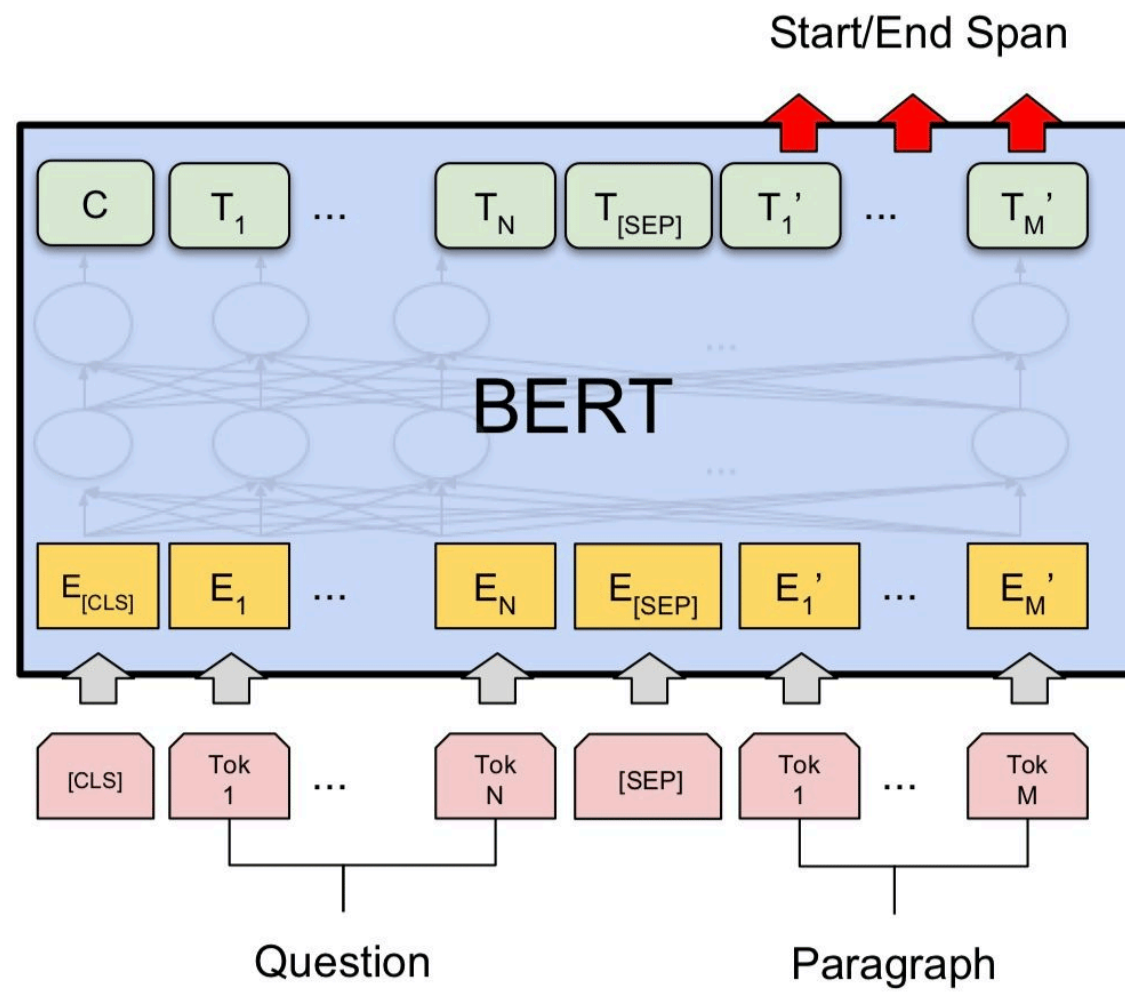
# Question Answering (QA)

# QA

**QA** is the task of **retrieving a span of text from a context** that is best suited to answer a question.

This task is extractive, and can be seen as information retrieval (more on that later).

# QA



# QA

The loss is the cross entropy over the output of the starting token and the ending one:

$$\mathcal{L}_{CE_{QA}} = \mathcal{L}_{CE_{start}} + \mathcal{L}_{CE_{end}}$$

# Natural Language Inference (NLI)



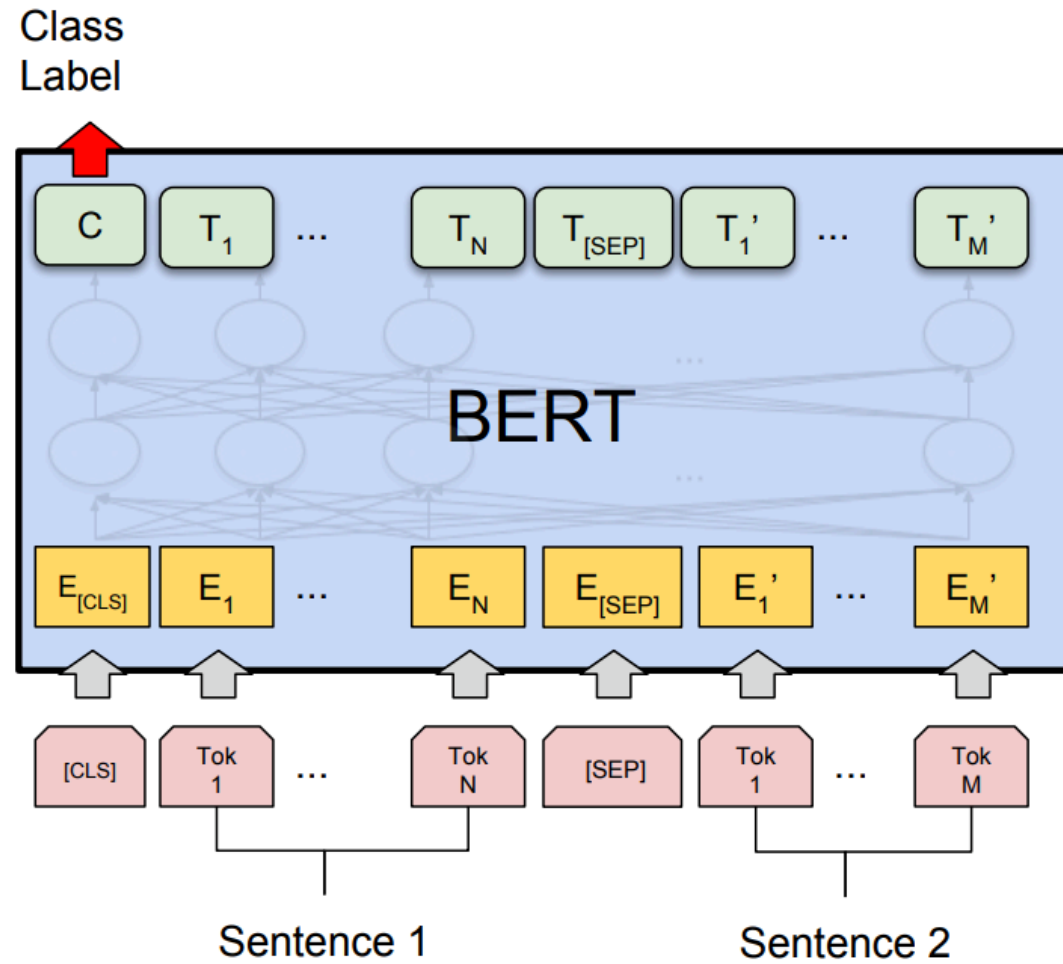
# NLI

**NLI** is the task of **determining whether a "hypothesis" is true (entailment), false (contradiction), or undetermined (neutral)** given a "premise".

# NLI

| Premise                                                            | Label         | Hypothesis                                                         |
|--------------------------------------------------------------------|---------------|--------------------------------------------------------------------|
| A man inspects the uniform of a figure in some East Asian country. | contradiction | The man is sleeping.                                               |
| An older and younger man smiling.                                  | neutral       | Two men are smiling and laughing at the cats playing on the floor. |
| A soccer game with multiple males playing.                         | entailment    | Some men are playing a sport.                                      |

# NLI



# NLI

The loss is simply the cross entropy or the divergence over the output of the **CLS** token and the true label.

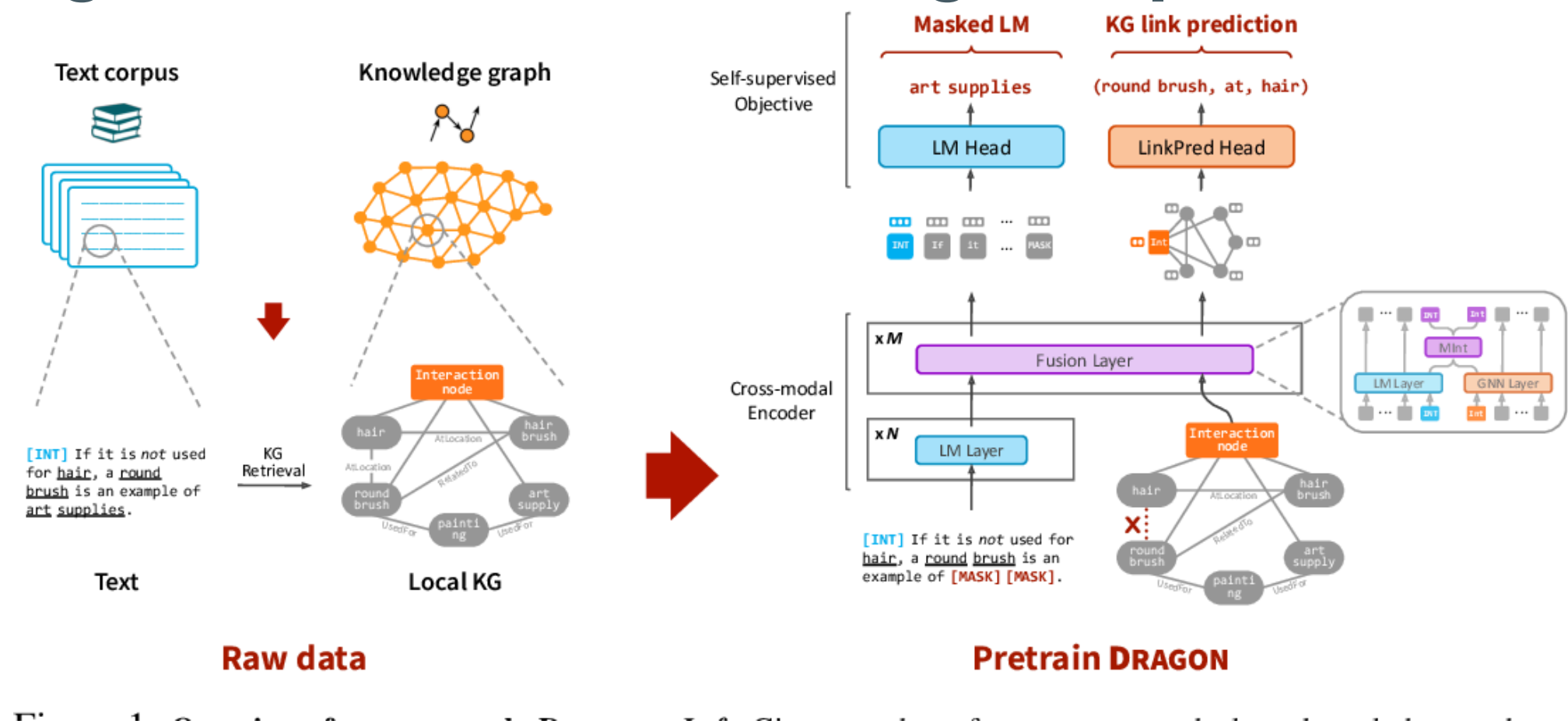
$$\mathcal{L}_{NLI} = \mathcal{L}_{CE_{CLS}}$$

We are trying to compress the information about both sentence in one **CLS** token via attention and decide about their relationship.

Is it possible to help the model inferring more information with less text data?

# Going Further: LM as Knowledge Graphs

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# Going Further: LM as Knowledge Graphs

|               | CSQA        | OBQA        | Riddle      | ARC         | CosmosQA    | HellaSwag   | PIQA        | SIQA        | aNLI        |
|---------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| RoBERTa [18]  | 68.7        | 64.9        | 60.7        | 43.0        | 80.5        | 82.3        | 79.4        | 75.9        | 82.7        |
| QAGNN [8]     | 73.4        | 67.8        | 67.0        | 44.4        | 80.7        | 82.6        | 79.6        | 75.7        | 83.0        |
| GreaseLM [9]  | 74.2        | 66.9        | 67.2        | 44.7        | 80.6        | 82.8        | 79.6        | 75.5        | 83.3        |
| DRAGON (Ours) | <b>76.0</b> | <b>72.0</b> | <b>71.3</b> | <b>48.6</b> | <b>82.3</b> | <b>85.2</b> | <b>81.1</b> | <b>76.8</b> | <b>84.0</b> |

1• Accuracy on downstream commonsense reasoning tasks. DRAGON consistently outperforms the existing

Improvements are mostly on dataset with few training examples and complicated examples (negations, non-verbal sentences, ...).

# Going Further: LM as Knowledge Graphs

This architecture *involves a KG ready to use beforehand and pre-training from scratch.*

How can we better **perform NLP task without having to retrain or fine-tune** a model?



# **Exploit LLMs capacities: Chain-of-thoughts & In context Learning**

# Exploit LLMs capacities

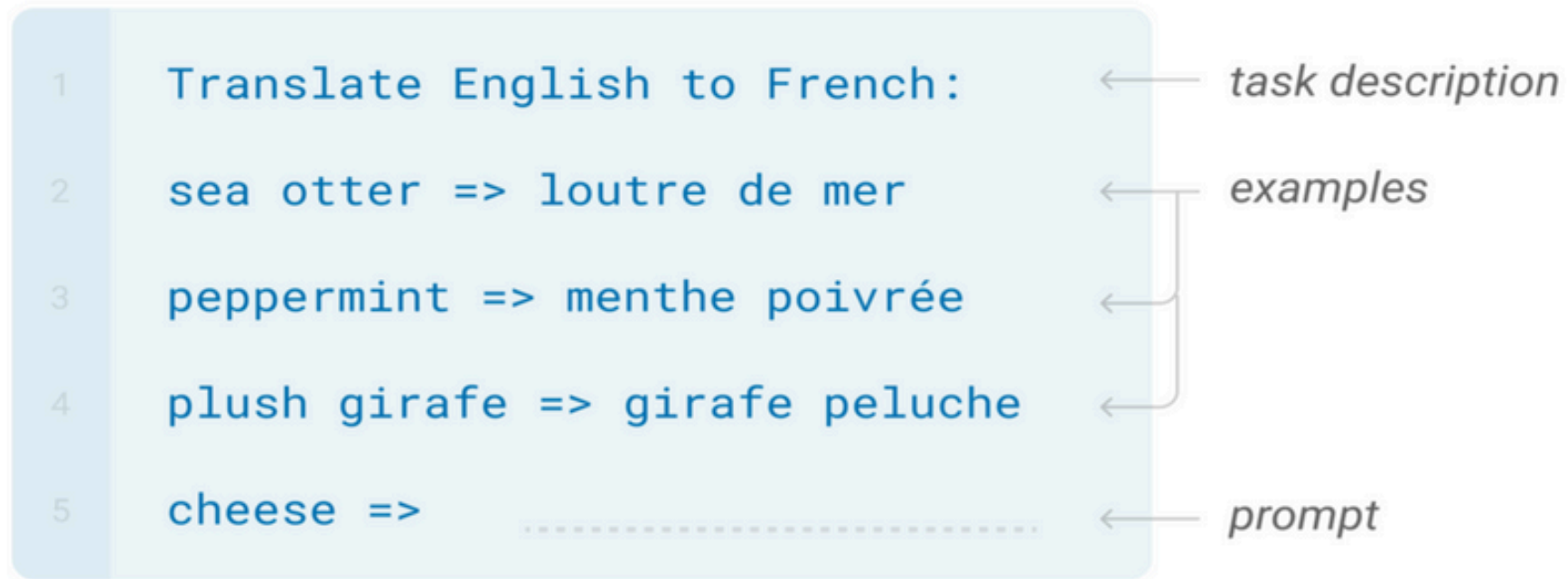
**ICL** enables LLMs to **learn new tasks** using natural language prompts **without explicit retraining or fine-tuning**.

The **efficacy** of ICL is **closely tied to** the model's **size**, training **data quality**, and **domain specificity**.

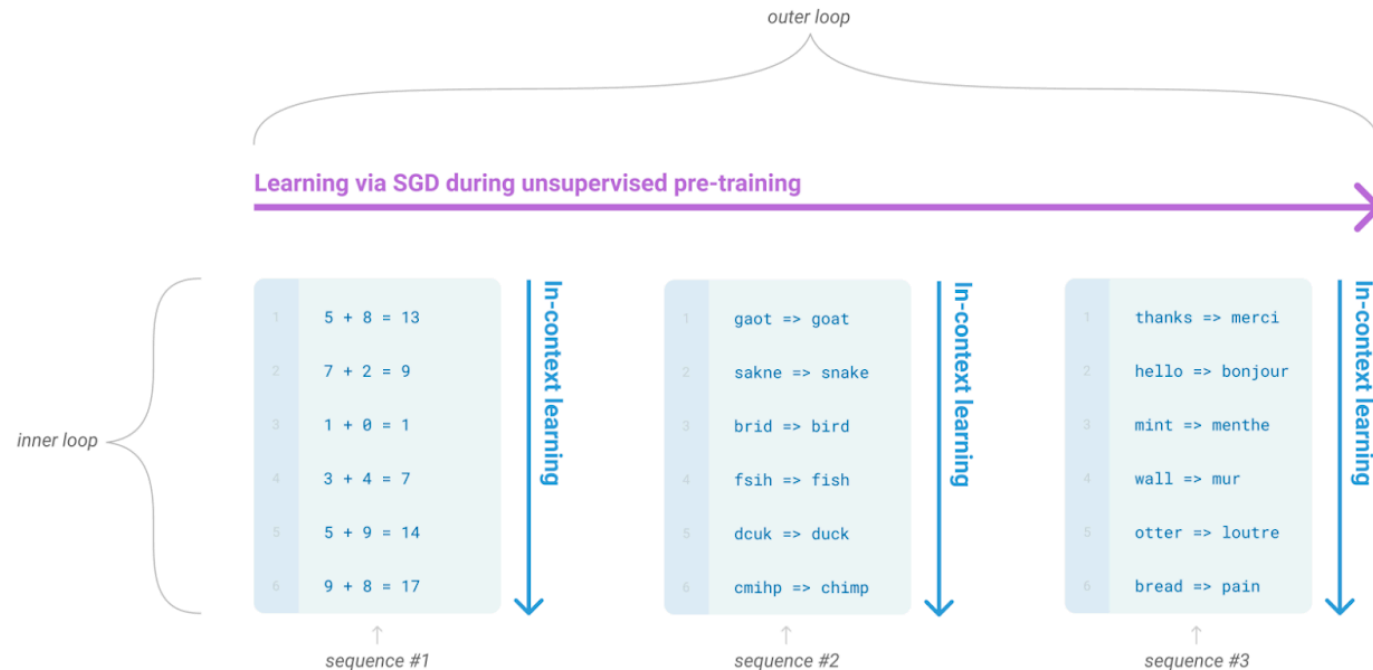
# Exploit LLMs capacities

## Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

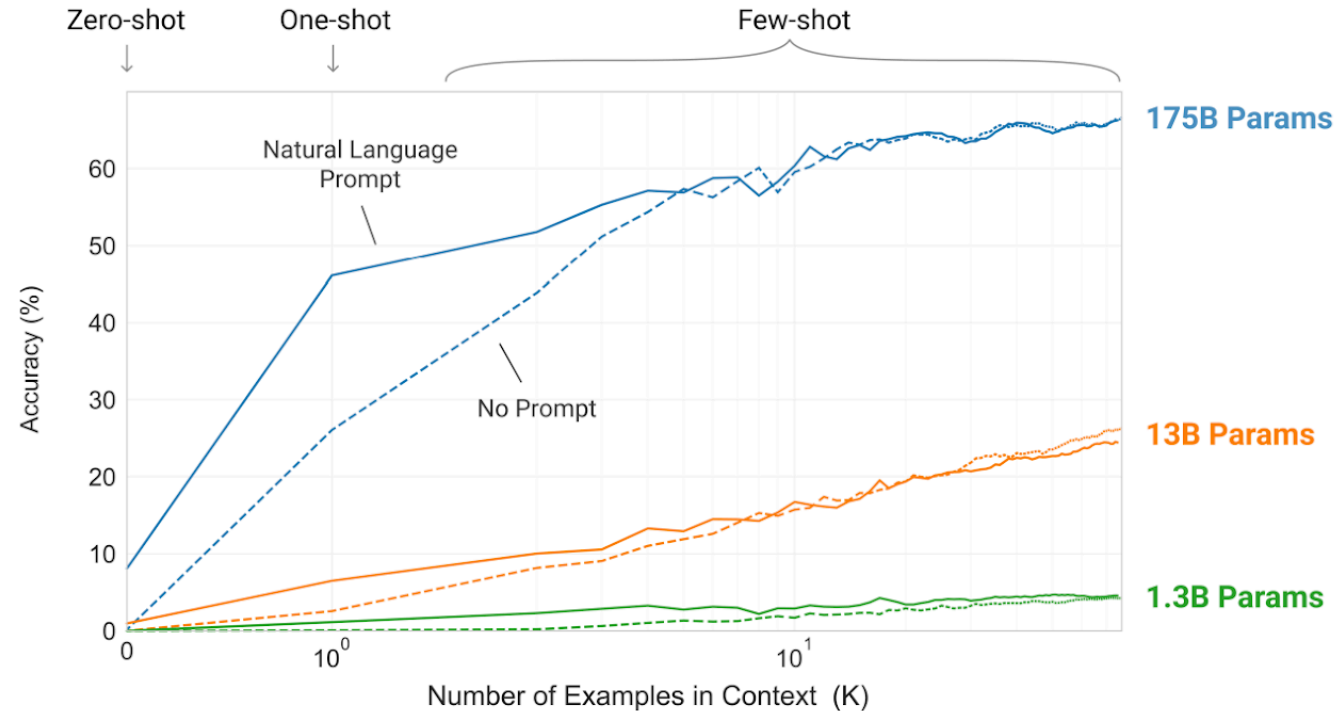


# Exploit LLMs capacities



**Figure 1.1: Language model meta-learning.** During unsupervised pre-training, a language model develops a broad set of skills and pattern recognition abilities. It then uses these abilities at inference time to rapidly adapt to or recognize the desired task. We use the term “in-context learning” to describe the inner loop of this process, which occurs within the forward-pass upon each sequence. The sequences in this diagram are not intended to be representative of the data a model would see during pre-training, but are intended to show that there are sometimes repeated sub-tasks embedded within a single sequence.

# Exploit LLMs capacities



**Figure 1.2: Larger models make increasingly efficient use of in-context information.** We show in-context learning performance on a simple task requiring the model to remove random symbols from a word, both with and without a natural language task description (see Sec. 3.9.2). The steeper “in-context learning curves” for large models demonstrate improved ability to learn a task from contextual information. We see qualitatively similar behavior across a wide range of tasks.

# Exploit LLMs capacities

(a) Few-shot

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The answer is 8. ✗

(b) Few-shot-CoT

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls.  $5 + 6 = 11$ . The answer is 11.

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A:

(Output) The juggler can juggle 16 balls. Half of the balls are golf balls. So there are  $16 / 2 = 8$  golf balls. Half of the golf balls are blue. So there are  $8 / 2 = 4$  blue golf balls. The answer is 4. ✓

(c) Zero-shot

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: The answer (arabic numerals) is

(Output) 8 ✗

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

# Questions?